

CASEPATH: AN AGENTIC, PROCESS-FIRST ARCHITECTURE FOR DETERMINING EVIDENCE REQUIREMENTS

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ABSTRACT

Insurance claims are a useful testbed for AI decision systems because the evidence needed to resolve a case depends on where that case lies in a rule-governed process. A single fact can activate a different branch, create new obligations, and change which documents are justified. Yet many LLM-based approaches bypass this structure and predict document checklists directly from case text. Such requests may look reasonable while being weakly grounded: the system can ask for the right-looking document for the wrong reason. We introduce CASEPATH, an agentic, process-first architecture in which the source-defined process—not the language model—controls evidence acquisition. Agents read authoritative sources, place the case on the applicable process path, identify active or unresolved requirements, and request only the evidence needed to satisfy them. As evidence arrives, the case state is updated and the required checklist changes accordingly. We test this dependency using matched cases that differ in one branch-defining fact. Across 27 held-out pairs, CASEPATH makes 15 false-positive document changes, versus at least 27 for every comparison system. This turns document prediction into evidence planning that is selective, traceable, and auditable.

1 INTRODUCTION

A customer reports a burglary. One stolen watch is separately scheduled under the insurance policy, which creates an obligation to establish its value. A purchase receipt or an expert valuation can do so. If the watch is not separately scheduled, that obligation disappears. If its status is unclear, the handler must first ask a question. Within the same claim category, a document can therefore be required, irrelevant, premature, or one of several sufficient alternatives.

Predicting a checklist directly leaves the reason for each request implicit. A model may have learned the applicable rule, or it may simply associate a familiar document with theft. CASEPATH expresses the dependency as an executable chain:

source → process node → obligation → required fact → evidence capability → document route. (1)

The model interprets source passages and assesses the case. A controller then derives requests from the active obligations. It asks a question when a condition is unresolved and makes no branch-specific request when the condition is false. Each request records its source, condition and purpose. As evidence arrives, the controller updates the case state and recomputes the plan (Figure 1). We test whether this order of operations makes evidence plans more selective when a case changes.

Study A measures whether a system changes the right requests after one case fact changes. It contains 36 matched pairs. On the 27 held-out pairs, CASEPATH makes 15 false-positive document changes, compared with 27 for Direct, 41 for Graph as context and 52 for Evidence-first. Against Direct, this is twelve fewer false-positive changes for one fewer correct change; signed-change F_1 is 0.667 versus 0.588. The advantage is concentrated in procedural branches, where the claim category gives little indication of the required document. Results by family also expose semantic errors that continue to produce wrong requests.

Study B evaluates complete plans on the 150-claim tenancy release. The registered evaluation encounters a native interface failure. Separate retrospective analyses measure literal-request agreement and evaluate plans at the model’s saved assessment. The latter isolates the effect of inherited process

Source rule: a scheduled valuable requires evidence of its value

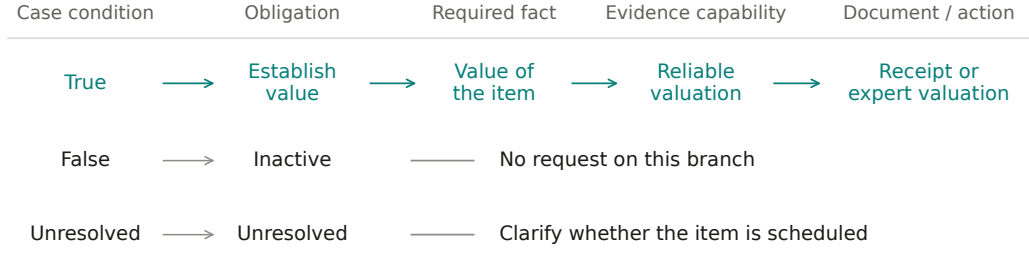


Figure 1: How the active branch determines a request. In this schematic example, a source condition makes an obligation active, inactive or unresolved. An active obligation specifies a fact to establish, an evidence capability and the document routes that can provide it. An unresolved condition calls for clarification. Section 2 defines each version’s route-selection and sufficiency rules.

scope. On the 90 protected-family cases, scope increases reference-chain precision from 0.318 to 0.732, with a conservative paired benefit of +0.103 after penalizing unavailable pairs. The volume result is plainer: without inherited scope the controller issues 853 requests to recover 376 valid ones, and with it 385 to recover 384. Critical-evidence recall is 0.645 against 0.627, so the reduction costs no coverage. Obligations that inherit the conditions of their enclosing process ask for less because what they drop was never justified.

We contribute a source-grounded representation of evidence requirements, an executable controller and evaluations of branch-induced changes and complete request lists. Compiled procedures and evidence ledgers already separate semantic judgement from control (Kim, 2026; Yu et al., 2026). We study how that separation affects evidence acquisition. The released benchmarks, outputs and evaluator allow readers to reproduce the measurements; the workbench exposes the source and obligation behind each request.

2 METHOD: DERIVING REQUESTS FROM PROCESS STATE

2.1 THE PLANNING OBJECT

The source specification K records process nodes and their conditions, obligations, required facts, evidence capabilities and document routes. A capability describes what the evidence must establish. A route is a subset of the closed document catalogue $\mathcal{D}$ that can establish it; a capability may have several alternative routes. Case state $s = (z, E)$ contains the assessment z of the source conditions and the recorded evidence E . The versioned planner returns documents, a next action and the chains that justify them:

$$(D_t, a_t, J_t) = \Pi_K(s_t), \quad s_{t+1} = U(s_t, e_{t+1}), \quad (2)$$

Here U admits an observation or correction and updates the state from which the next plan is computed. The construction gives two guarantees. An unresolved condition can produce a question but cannot trigger a branch-specific document request (non-prematurity). Every document in D_t comes from an active chain containing an obligation, a fact, a capability and a route (provenance closure).

2.2 MODEL JUDGEMENTS AND CONTROLLER DECISIONS

Model calls perform the semantic work. During preparation, fixed roles extract propositions anchored to source quotations, synthesise the macro process, refine propositions into guarded evidence rules and map capabilities to the document catalogue. At inference time, the interpreter judges whether each guard holds for the case. These roles return structured judgements. Code activates obligations, projects requests, selects routes and prioritises the next action, recording the justification for each choice (Appendix A). The model can interpret a passage or assess a condition; document requests must pass through the controller.

2.3 STUDY A: GUARDED RULES OVER A CLOSED CATALOGUE

We compiled the frozen Study A specification from 197 exact passages of Swiss insurance law and public household-insurance policies and forms. The sources are Fedlex, CSS, AXA, Generali, Simpego and the Swiss Insurance Association; the release records their URLs and content hashes. Extraction produced 308 operational propositions. A deterministic check requires an evidence rule or an explicit non-evidentiary classification for each proposition. The resulting representation contains 98 evidence rules, 13 non-evidentiary classifications, 87 case guards and 33 document types. This completeness check covers the extracted inventory.

Each rule r has a local applicability guard g_r , required capabilities $\mathcal{C}_r$ and routes $\mathcal{R}_c$ for each capability c . For case x , the interpreter assigns $z_g(x) \in \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ to guard g . A true or false verdict requires an exact quotation from that case; otherwise the verdict becomes unresolved. Missing evidence also leaves a guard unresolved. Each guard has its own model call, although case packets are batched for execution, so individual branch decisions remain auditable. Let $A(x)$ contain the unconditional rules and those whose guards are true. Let $\mathcal{R}_g^{\text{conf}}$ contain the mapped confirmation routes for a guard that is true in the narrative but has not been established by admitted evidence. Given held document types H , the planner returns

$$D(x, H) = \left[\bigcup_{r \in A(x)} \bigcup_{c \in \mathcal{C}_r} \bigcup_{\rho \in \mathcal{R}_c} \rho \cup \bigcup_{g \text{ true, unconfirmed}} \bigcup_{\rho \in \mathcal{R}_g^{\text{conf}}} \rho \right] \setminus H. \quad (3)$$

Thus a false guard adds no demand, an unresolved guard produces a question, and the accepted verdicts fully determine the plan. Each request retains its rule, fact, must-show statement and source spans.

Equation 3 also defines this version’s limits. V5 unions the mapped routes: it does not choose one sufficient alternative or assess a held document’s adequacy for a particular capability. Activation depends on local rule guards. Changing macro-process edges while keeping the rules and verdicts fixed leaves D unchanged. In Study A, an “active path” is this case-conditioned applicability structure; it does not require traversing every graph edge.

2.4 STUDY B: APPLICABILITY, ACQUISITION AND ADEQUACY

Study B deterministically compiles three public tenancy templates into a representation shared by all arms. It uses no model-based source induction. The case assessment distinguishes predicate truth, document presence, individual adequacy and joint adequacy. Activation follows

$$q_s(z) = \phi_s(z) \wedge_3 J_s(\{q_p(z)\}_{p \in \text{pa}(s)}), \quad a_o(z) = q_{\text{scope}(o)}(z) \wedge_3 \phi_o(z), \quad b_o(z) = a_o(z) \wedge_3 \psi_o(z), \quad (4)$$

where J_s is the declared all/any join over parent scopes. Roots use the empty all-join, which is true. The three-valued conjunction $\wedge_3$ is false if any input is false, true if all are true and unresolved otherwise. An obligation’s applicability a_o therefore depends on both its enclosing scope and its own condition. The separate acquisition permission ψ_o must also hold: only $b_o = \mathbf{T}$ permits an immediate request.

A route ρ satisfies capability c only when the required evidence is present and adequate and the joint assessment is sufficient. Denote this state by $\rho_c(E)$. Missing or inadequate documents form $M_{c\rho}$; unknown presence or adequacy creates review items $V_{c\rho}$. The controller selects

$$\rho_c^* = \arg \min_{\rho \in \mathcal{R}_c}^{\text{lex}} (\mathbf{1}\{\rho_c(E) \neq \mathbf{T}\}, |M_{c\rho}|, |V_{c\rho}|, \text{sort}(M_{c\rho}), \text{id}(\rho)), \quad (5)$$

independently for each capability, then requests $\bigcup_c M_{c\rho_c^*}$ over eligible obligations. This is a local route choice, not a global minimisation. Inactive obligations produce no demand. Unresolved eligibility produces conditional candidates and questions. If no route exists, the controller records an evidence gap. Its action priority is to resolve questions, review evidence, request a document, review an active gap and, finally, choose an eligible substantive action.

This controller adds sufficiency semantics to the Study A planning object. It is a separate runtime, and the experiments do not compare its performance with V5 or treat it as a repair of V5 (Appendix B).

3 STUDY A: CHANGING THE ACTIVE BRANCH

Case pairs and reference changes. We selected 9 branch concepts for which changing a fact changes at least one evidence requirement in the sources: a scheduled valuable at or above the policy threshold, a stolen bicycle, SIM-card theft with misuse, an identifiable third-party causer, a formal expert procedure, a repair above the consent threshold, a written demand with a forfeiture period (*Nachfrist*), recovery of stolen goods and insurer declination. Each concept appears in four neutral household-theft contexts, giving 36 pairs and 72 case units. The two cases in a pair differ in one statement. For the watch example, “No stolen item is separately scheduled in the policy with an insured value of CHF 1,000 or more” becomes “A stolen watch is separately scheduled in the policy with an insured value of CHF 4,000”.

Before running any system, we authored a reference contract from the same sources to specify the acceptable signed changes. It preserves alternatives such as a receipt or an expert valuation. Both amendments made before the held-out evaluation remain available with the earlier benchmark versions. One context per concept supplies 9 development pairs (18 units). The other three contexts, 27 pairs (54 units), stayed sealed until the method, benchmark, analysis code and gate were hash-frozen, and were read once. This split tests known rules in new contexts. It does not test transfer to unseen rules or policies.

Controls for shortcut explanations. We tested whether static predictions, request volume or branch-independent changes could explain an apparent gain (Kaushik & Lipton, 2018; Sugawara et al., 2018; McCoy et al., 2019; Bowman & Dahl, 2021; Yuan et al., 2024). A constant signed-set predictor fitted on eight branch families and tested on the ninth reaches maximum held-out micro- F_1 0.0. The nine target target signatures are distinct, with 3.17 bits of entropy over 12 signed document atoms.

The wrong-pairing control keeps CASEPATH’s outputs fixed and reassigns them to other branch families using 8 cyclic shifts and 5,000 seeded derangements. A score that survives this reassignment would not require branch-specific changes. We also fixed eight acceptance conditions before the single held-out execution (Table 4), including a strong requirement for broad superiority.

Scoring. For a pair (x^-, x^+) , the predicted signed change is

$$\Delta(D) = \{+d : d \in D(x^+) \setminus D(x^-)\} \cup \{-d : d \in D(x^-) \setminus D(x^+)\}. \quad (6)$$

An addition enters with a positive sign and a withdrawal with a negative sign. The route-aware scorer compares Δ with each acceptable reference realisation and selects the best under a frozen lexicographic rule (Appendix A). Requesting both alternatives when one suffices does not receive full credit. We pool true positives, predictions and reference changes across the 27 pairs to compute micro precision, recall and F_1 . A spurious change is an unjustified addition or withdrawal; a missed change is required but absent. An exact pair reproduces the entire signed set. These metrics concern changes between checklists, so a request shared by both cases does not affect them. We estimate uncertainty with 10,000 bootstrap draws of whole families and test paired differences with the exact $2^9 = 512$ family sign-flip test, applying Holm adjustment.

Comparators and computation. Direct predicts documents from the case and source material. Graph as context also receives the induced process and evidence-rule representation, then generates the document set in one step. Evidence-first derives evidence needs, retrieves twelve passages by keyword matching and predicts documents. Every arm uses model setting `openai/gpt-5.6-terra`, the same provider restriction, temperature zero, source snapshot, catalogue and case texts. The held-out receipts record 54, 54, 108 and 87 calls, respectively. The comparisons hold these inputs and model conditions fixed, but allow different amounts of computation (Appendix A).

4 STUDY A: FEWER UNJUSTIFIED CHECKLIST CHANGES

Selectivity and recall. CASEPATH predicts 39 signed changes against 33 reference changes: 24 are correct, 15 are spurious and 9 are missed (Table 1). Direct predicts 52 changes, with 25 correct and 27 spurious; Graph as context predicts 72, with 31 correct and 41 spurious; and Evidence-first

Method	Signed changes			Agreement	
	Correct	Spurious	Missed	Exact pairs	F_1 [95% CI]
Direct	25	27	8	9/27	0.588 [0.404, 0.759]
Graph as context	31	41	2	4/27	0.590 [0.471, 0.716]
Evidence-first	16	52	17	2/27	0.317 [0.239, 0.400]
CASEPATH	24	15	9	12/27	0.667 [0.435, 0.880]

Table 1: Signed-change results on 27 held-out pairs with 33 reference changes. Correct, spurious and missed count signed changes; Exact counts pairs whose full signed set is reproduced. The F_1 interval is a 95% family-cluster bootstrap interval. These counts describe changes between two checklists, not requests for a complete claim.

predicts 68, with 16 correct and 52 spurious. At recall 0.727, CASEPATH makes 44.4%, 63.4% and 71.2% fewer spurious changes than these alternatives. It reproduces the whole signed set on 12 of 27 pairs, compared with 9, 4 and 2.

Graph as context recovers the most reference changes, 31 of 33, but also produces 41 spurious changes. Supplying a graph helps recall in this comparison; enforcing its rules restricts the requests. Against Direct, CASEPATH makes twelve fewer false-positive changes while missing one additional correct change.

A retrospective error partition identifies documents outside the benchmark’s branch-governed set. They account for 9 of CASEPATH’s 15 spurious changes and all 27, 41 and 52 comparator errors. The controller reduces this checklist drift without eliminating it. The signed-change references do not identify which baseline documents would be required in every case, so this partition cannot establish that those documents are universally necessary.

Uncertainty and the preregistered gate. The family-cluster interval for CASEPATH’s F_1 of 0.667 is [0.435, 0.880]; Direct and Graph as context have intervals [0.404, 0.759] and [0.471, 0.716]. The paired-difference intervals against those two systems, $[-0.253, +0.420]$ and $[-0.199, +0.333]$, include zero. Only the interval against Evidence-first excludes zero ($[+0.110, +0.565]$). Three of the eight preregistered gate conditions fail (Table 4): the margins +0.078 and +0.076 are below 0.10, and precision 0.615 is below 0.70. The Holm-adjusted family-swap p -values are 0.797, 0.797 and 0.129. The lower error count supports selectivity on these pairs, while the preregistered criterion for broad superiority remains unmet.

Dependence on the branch. Reassigning the saved outputs to the wrong branch families reduces signed-change F_1 from 0.667 to at most 0.083 over 5,008 evaluated reassignments (mean 0.011; add-one randomisation $p = 1.996 \times 10^{-4}$). Thus the document changes depend on which branch was changed. This test does not isolate the representation as their cause, since a family-specific lookup could also pass.

The records attribute all 39 of 39 predicted changes, including errors, to a changed guard. The frozen representation also contains a source chain for 27 of 33 reference changes (81.8%). These checks establish traceability and representational coverage. A trace can contain an incorrect mapping, and an available route need not appear in the emitted plan.

What the family results explain. CASEPATH is exact in all three held-out contexts of four concepts: bicycle, declination, recovery and *Nachfrist*. It reaches 0.667 on scheduled valuables and SIM-card theft, 0.500 on repair consent, and zero on third-party causation and the formal expert procedure (Figure 2). The failures recur across contexts. They arise when the planner requests an alternative route in addition to a sufficient one, a condition has the wrong semantic scope, or an adjacent procedural guard activates the wrong confirmation document.

The comparators do well when a plausible claimant document supplies the answer, as with third-party causation and repair consent. They struggle when the required change concerns a procedural artefact, as with declination and *Nachfrist*. Appendix A reports every concept’s score. These paired rule units leave one question open: a document wrongly included or omitted in both checklists produces no signed error. Study B therefore evaluates complete plans for whole claims.

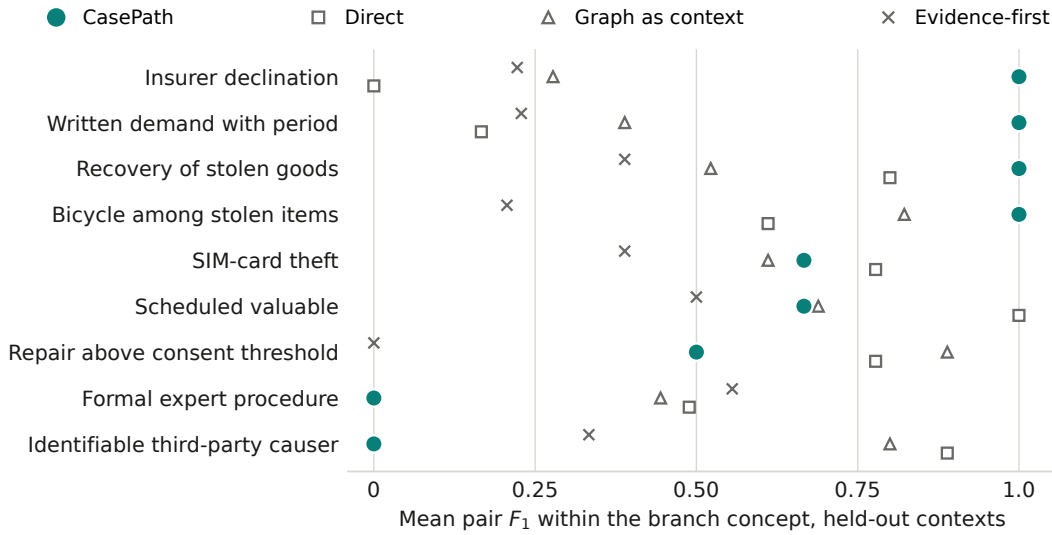


Figure 2: Mean pair F_1 for each branch concept in the held-out contexts. The CASEPATH marker is compared with squares for Direct, triangles for Graph as context and crosses for Evidence-first. Small vertical offsets separate coincident values. CASEPATH is exact on four concepts where the comparators are not, and scores zero on two where they succeed. Concepts are ordered by CASEPATH’s score.

5 STUDY B: COMPLETE CLAIMS AND INHERITED PROCESS SCOPE

Corpus and matched conditions. Study B includes all 150 released synthetic tenancy claims, covering 28 families in three domains: defects, mould and heating; termination; and rent increase. The development split contains 60 claims in 11 families. The protected split contains 90 claims in 17 disjoint families. Five learned pipelines share the compiled public representation, observable packet, source access and model alias. Each has the same budget of 2 calls per cell, medium reasoning and 4,096 completion tokens per call, with no tools or resampling. CASEPATH produces an assessment that code compiles into an artefact. Direct, Document-first, Process-context and Rule-first generate the artefact themselves. Compiled-equivalent and local-scope controls reuse CASEPATH’s assessments. The comparison therefore holds the available information and budget fixed across complete pipelines.

Registered evaluation and retrospective analyses. All 558 generated artefacts fail the original native interface because the reference scenario does not bind the compiler’s condition names. Under the frozen policy, all 1,050 cells receive adverse penalties, including 492 failed or blocked outputs. The primary result meets 0 of 12 protected-split practical targets. We preserve that report and every failure. Two separately specified retrospective analyses examine what the saved outputs can establish: agreement between literal requests and reference sets, and the effect of inherited scope on a plan evaluated at its recorded assessment. These analyses leave the registered result unchanged.

Request quality across the complete comparison. The first analysis applies the original reference and counting functions to literal immediate requests, without interpreting guards. All 558 generated lists can be scored. Failed outputs retain zero quality scores and an unnecessary-fraction penalty of one. Table 2 reports every arm. On protected families, CASEPATH produces 66 lists and reaches failure-penalized F_1 0.677. Process-context, the strongest comparator, reaches 0.458. Output availability contributes to this difference. The conventional paired advantage is +0.219; a conservative contrast that assigns -1 whenever either endpoint is unobserved gives -0.673. CASEPATH has better realized output under the budget, but these comparisons do not establish reliable paired superiority over the learned alternatives.

Pipeline	Generated lists		Checklist $F_1 \uparrow$		Unnecessary $\downarrow$	
	Dev.	Prot.	Dev.	Prot.	Dev.	Prot.
CASEPATH	44	66	0.657	0.677	0.296	0.268
Direct	19	30	0.295	0.310	0.687	0.688
Document-first	17	24	0.247	0.249	0.732	0.737
Process-context	27	43	0.407	0.458	0.570	0.538
Rule-first	28	42	0.433	0.450	0.545	0.540
<i>Dependent controls: same recorded CasePath assessments</i>						
Compiled equivalent	44	66	0.657	0.677	0.296	0.268
Local-scope ablation	44	64	0.420	0.411	0.669	0.682

Table 2: Agreement of literal request lists with reference sets across the complete corpus. This is a retrospective analysis. Domains receive equal weight, as do families within a domain and cases within a family. Failed outputs remain in the denominator; their “Unnecessary” penalty is one. The dependent controls use the same recorded assessments.

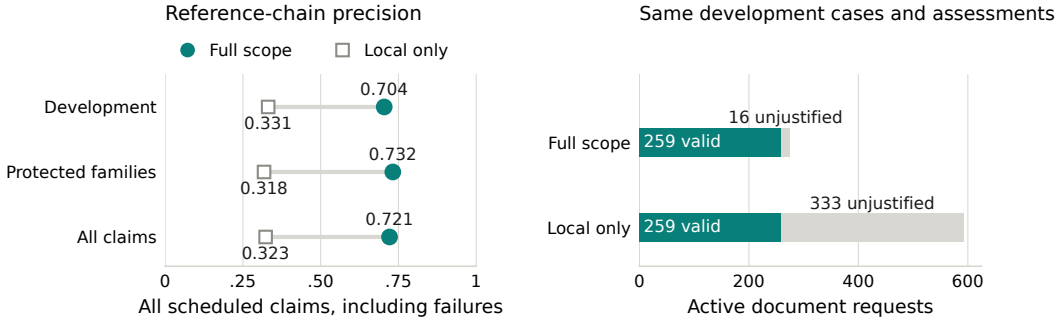


Figure 3: Effect of inherited scope at fixed assessment. Left: reference-chain precision in the retrospective current-case evaluation, with every scheduled case and failure penalty included. Right: on the same 44 completed development cases, both controllers retain 259 valid requests, while inherited scope reduces total requests from 592 to 275. The assessment and source representation are fixed.

Evaluation at the recorded assessment. The second analysis binds each saved artefact to its own authenticated assessment. Conditions known to be true activate their entities; false conditions leave them inactive. Unresolved entities remain recorded as unresolved and are not asserted active. This creates a snapshot of the current case that can be scored using the model’s recorded beliefs. The original requests and all failed cells are retained. Only CASEPATH and its two dependent controls have the saved assessments required for this analysis. Every available snapshot scores successfully, and none excludes a literal immediate request. The snapshot evaluates current-case structure; it cannot test the program’s response to hypothetical changes in branch conditions (Appendix E).

Scope removes unjustified demand. The intervention in Figure 3 requires each obligation to satisfy its enclosing process conditions as well as its local guard. On protected families, this increases native valid-chain precision from 0.318 to 0.732. Even after penalizing unavailable pairs, the conservative paired benefit is positive at +0.103. The unnecessary fraction falls from 0.682 to 0.268. Precision and unnecessary fraction describe complementary parts of the same requests, so they count as one effect.

On protected families the same intervention is starker in volume: the ablation issues 853 requests to recover 376 valid ones, while CASEPATH issues 385 to recover 384. Coverage does not pay for that reduction. Critical-evidence recall is 0.645 against 0.627 and required-node recall 0.544 against 0.526, while required-node precision separates by more than a factor of two, 0.536 against 0.256. Inherited scope is therefore not a precision–recall trade: it withdraws demands that no active obligation supported, and the obligations that were supported survive. On the same 44 completed development cases the pattern repeats exactly, with both controllers recovering 259 valid requests at identical recall (0.635) while total requests fall from 592 to 275. The compiled-equivalent control

matches CASEPATH exactly. This intervention tests inherited domain applicability, without isolating arbitrary graph topology. Native chain matching compares the decision, fact, capability and document links with the reference. It does not establish exact-source entailment: provenance-chain inheritance scores zero for both controllers.

The protected-family label denotes the original target-custody split. Product development had exposed the observable inputs, and the missing producer I/O audit prevents a retrospective claim of target non-access. The targets are now released. These descriptive finite-corpus analyses therefore provide no fresh hidden confirmation. The study cost is \$89.06, or \$89.66 including earlier activity; the retrospective analyses require no additional provider calls. Appendices D and E give the full endpoints, raw counts and paired contrasts.

6 RELATED WORK

Executable rules and case state. CASEPATH builds on the established separation of rules from the state to which they apply. LegalRuleML records norms and their source associations (OASIS LegalRuleML Technical Committee, 2021), and Catala makes legislative computation executable (Merigoux et al., 2021). DMN formalises decisions; CMMN represents case work activated by guards (Object Management Group, 2024; 2016). Guard–Stage–Milestone models and DCR reference models of law connect activation driven by data to compliance (Vaculín et al., 2011; López et al., 2020). Proactive enforcement derives actions from active obligations (Basin et al., 2024), while insurance software conditions document lists on service definitions (ServiceNow, 2026). We combine this approach with language-model assessments and measure the resulting evidence-planning errors.

Procedural graphs and workflow agents. PAGED extracts procedural graphs from documents (Du et al., 2024). BREX evaluates business-rule flows with executable grounding (Yang et al., 2026), and Guideline2Graph compiles clinical guidelines into decision graphs that retain provenance (Kilic et al., 2026). Graus (2026) evaluate structured representations for legal decision models; CANONIC treats governance as compilation (Hadley, 2026). FlowBench and WorFBench evaluate workflow-guided planning and workflow generation (Xiao et al., 2024; Qiao et al., 2025). Procedural Graphs place an agent at an active node (Lu et al., 2026), while ReAct interleaves reasoning and action (Yao et al., 2023). Our experiments ask whether changing a case fact changes exactly the evidence justified by the affected branch.

Evidence obligations and acquisition. Kim (2026) provide the closest evidence-ledger precedent: branch conditions activate knowledge obligations, a model judges semantic warrant and code manages closure. Yu et al. (2026) compile SOP constraints into verification routines with evidence-bearing returns and an active execution frame. We study these principles for document planning, where applicability and acquisition permission can differ, a present document may be inadequate, and several documents may need joint assessment. The representation retains alternative evidence routes, and the benchmark measures their effect on requests. Neither prior system is reproduced as an experimental arm, so our comparisons do not establish superiority over them.

Grounding and information seeking. GraphCompliance aligns policy and context graphs for regulatory assessment (Chung et al., 2025). CLER scores litigation evidence lists and the facts they prove (Bi et al., 2026). InfoGatherer uses evidential uncertainty to select retrieval and questions (Taranukhin et al., 2026). Verified quotations and chain-of-verification expose the support for a response (Menick et al., 2022; Dhuliawala et al., 2024). API-Bank and BFCL evaluate function selection in tool use (Li et al., 2023; Patil et al., 2025). Our measured object is the justification for requesting evidence in a particular case.

Evaluation validity. Work on partial-input baselines, annotation artefacts and shortcut learning motivates our input-free and wrong-pairing controls (Kaushik & Lipton, 2018; Poliak et al., 2018; Gururangan et al., 2018; McCoy et al., 2019; Sugawara et al., 2018; Geirhos et al., 2020; Bowman & Dahl, 2021; Yuan et al., 2024). We authored Study A’s references from the sources. No model adjudicated them, which avoids the preference leakage that can arise when the judged and judging models coincide (Li et al., 2026). Evaluation-validity research (Thomas et al., 2026) and critiques of legal benchmarks focused on outcome prediction instead of practitioners’ work (Medvedeva &

McBride, 2023; Lou & Shin, 2026) motivate our use of signed document changes and emitted requests as outcomes.

7 RESEARCH AND PRODUCT CORRESPONDENCE

The Study A product uses the experiment’s guard interpreter and planner. Its hash-bound method pack contains the frozen representation, propositions, source snapshot, method manifest, benchmark identity and model configuration. Clients cannot supply rules, mappings or targets; the endpoint rejects hash drift. The interface exposes the guard states and justification chains alongside questions, gaps, requests, exact source spans and the next action.

To check this correspondence, we replayed all 72 benchmark units through the product with their recorded guard decisions. Every unit reproduced the research document set, guard states and next action, as did all 36 signed pair changes. Every displayed quotation was an exact substring of its source passage. This verifies implementation identity conditional on the recorded assessments; fresh model agreement and hosted deployment identity require separate checks.

The workbench organises Facts, Orchestration, Source Integrity, Process, Evidence, and Audit and Readiness around one case record. Admitting a new observation recomputes the plan through Equation 2. The release contains the benchmark versions, predictions, receipts, analysis code and parity verifier, together with Study B’s corpus, producer, evaluator, contract and every recorded cell. The registered primary failure and the separate request-only diagnostic reproduce offline from the exact released reference containers. Reference targets, authored teaching states and model predictions appear separately; the first two supply no additional benchmark evidence.

8 DISCUSSION AND LIMITATIONS

Process control helps a system explain why a request belongs in the plan. In Study A, it reduces false-positive changes while recovering nearly as many correct changes as Direct. The advantage is largest on procedural branches, where the applicable rule calls for a particular notice or confirmation that a familiar claimant document cannot replace. Study B isolates a related requirement: an obligation must inherit the conditions of its enclosing process as well as satisfy its local guard. A system builder can inspect these dependencies and test their effect on requests.

The studies support these conclusions within defined settings. Study A covers known concepts in new contexts, with reference additions but no justified withdrawals. Computation differs across pipelines, and the preregistered broad-superiority gate fails. Its references reflect author judgements about which documents establish a fact, without expert consensus. Deployment would also require selecting the governing policy; the study combines rules from several insurers.

Study B uses complete synthetic claims and matched budgets. The native interface failure leaves counterfactual graph correctness unvalidated. Its retrospective current-case analysis isolates scope within dependent controls, while the broader request comparison measures realized output including failures. Neither study measures customer effort, expert agreement, autonomous acquisition or performance in another jurisdiction.

9 CONCLUSION

CASEPATH derives document demands from source conditions, active obligations and the facts that evidence must establish. The controlled comparison finds fewer unjustified checklist changes. The current-case intervention shows that inherited scope removes unjustified requests at no cost in coverage: on protected families it cuts requests from 853 to 385 while critical-evidence recall holds at 0.645 against 0.627. These findings support deriving evidence demands from explicit obligations evaluated against the case state. Both the resulting requests and the process semantics need validation. The released benchmarks, controller, traces and offline analyses make these checks reproducible.

REPRODUCIBILITY STATEMENT

For Study A, the release includes the frozen source snapshot, propositions, rules, document map, interpreter and planner, along with every benchmark version, prediction, provider receipt, analysis script, preregistered gate and product replay verifier. For Study B, it includes the corpus, compiled representation, producer, evaluator, analysis contract and every recorded cell, including failures. Scripts generate each numerical result in the paper and record its source-file hash and JSON pointer. The release commands recompute Study A and all three Study B analyses from the saved outputs, comparing them with the preserved measurements without network or provider access (Appendices A and D). Regenerating the model outputs would require provider access and would not be deterministic.

ETHICS STATEMENT

The studies use public-source snapshots and synthetic cases containing no personal data. CASEPATH is a research architecture for evidence planning; it provides neither legal advice nor automated coverage decisions. Use in practice would require policy and jurisdiction scoping, professional review, privacy controls, and procedures for correction and appeal.

AI USE STATEMENT

We used generative AI in hypothesis formulation, methodological critique, literature discovery, implementation and testing, synthetic-case construction, diagnostic interpretation, and the drafting and editing of text and figures. The reproducibility materials identify actual executions, engineering fixtures, archived reports and author judgements, and retain failed results and version differences. The authors take responsibility for the final claims, citations, measurements and artefacts.

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A STUDY A: BENCHMARK AND ANALYSIS DETAILS

Branch concepts. The 9 concepts cover a scheduled valuable at or above the source-defined threshold; a stolen bicycle or e-bike; SIM-card theft with a misuse claim; an identifiable third-party causer; selection of formal expert proceedings; contemplated repairs above the consent threshold; a written demand with a forfeiture warning and period; recovery of, or news about, stolen items; and insurer declination. Each appears in four neutral contexts. Only the branch statement changes within a pair.

Signed-change scoring. For pair i , the scorer compares prediction $P_i = \Delta(D_i)$ with every acceptable signed reference realisation $G \in \mathcal{G}_i$. It selects the maximum of the lexicographic tuple $(F_1(P_i, G), |P_i \cap G|, \text{Prec}(P_i, G), \text{Rec}(P_i, G), -|G|, \text{sort}(G))$. The final component supplies the frozen deterministic tie-break. Precision and recall are zero when their denominators are empty; F_1 is one when both the prediction and reference are empty. Micro scores pool counts for the selected realisations across 27 pairs. Family scores average pair F_1 within each of the nine concepts.

For each of 10,000 bootstrap draws, we sample nine whole families with replacement using seed 20260917. Each selected family retains all three of its held-out contexts. The exact family-swap test enumerates 512 sign assignments of the nine family mean differences. Its three one-sided p -values, 0.436, 0.398 and 0.043, become 0.797, 0.797 and 0.129 after Holm adjustment. The wrong-pairing diagnostic uses 8 cyclic shifts and 5,000 sampled derangements, with add-one score $(1 + N_{\geq})/(1 + 5,008)$. Its reported maximum covers the evaluated reassignments only.

Results by concept. Table 3 gives held-out mean pair F_1 for every concept and arm.

Branch concept	Direct	Graph ctx.	Evid.-first	CASEPATH
Bicycle among stolen items	0.611	0.822	0.206	1.000
Insurer declination	0.000	0.278	0.222	1.000
Formal expert procedure	0.489	0.444	0.556	0.000
Recovery of stolen goods	0.800	0.522	0.389	1.000
Written demand with period (Nachfrist)	0.167	0.389	0.229	1.000
Repair above consent threshold	0.778	0.889	0.000	0.500
Scheduled valuable	1.000	0.689	0.500	0.667
SIM-card theft	0.778	0.611	0.389	0.667
Identifiable third-party causer	0.889	0.800	0.333	0.000

Table 3: Held-out mean pair F_1 for each branch concept.

Preregistered gate. We fixed the eight conditions in Table 4 before the single held-out evaluation. The table records each condition’s outcome.

Preregistered held-out gate	Result
F_1 margin ≥ 0.10 against every comparator	fail
Holm-adjusted family-swap $p \leq 0.05$ against every comparator	fail
Signed-change precision ≥ 0.70	fail
Predicted/gold change ratio in $[0.75, 1.50]$	pass
Predicted changes attributed to a changed guard ≥ 0.90	pass
Reference changes reachable through a source chain ≥ 0.80	pass
Correct pairing above the wrong-pairing 97.5th percentile	pass
No requested document without a source chain	pass

Table 4: The eight preregistered conditions and their outcomes. Provenance, volume and pairing conditions pass; broad-superiority conditions fail.

Freeze and split handling. The method freeze `casepath.theft-method-freeze/5.0.0` binds the source bytes, benchmark v3 and both amendments, proposition-complete refinement, document mapping, interpreter V4 and planner V3. It also binds the model, provider and temperature settings, analysis code, development split, matched comparators’ raw predictions and focused method tests. Development used 9 pairs (18 units), one context per concept. Its predeclared admission checks authorised one read of the remaining 27 pairs (54 units) under a separate hash-bound execution freeze. The method remained locked after that read; no scientific file was adapted to the held-out outcomes.

Calls and costs. Table 5 reports the held-out provider receipts for each arm.

Arm	Calls	Prompt tokens	Completion tokens	Max completion	Cost (USD)
Direct	54	2,900,625	45,678	14,000	4.83
Graph as context	54	5,345,421	42,663	14,000	8.41
Evidence-first	108	259,890	66,450	14,000/18,000	1.06
CASEPATH	87	1,005,211	243,891	10,000	5.44

Table 5: Held-out calls and provider-reported charges for `openai/gpt-5.6-terra` at temperature 0, with one logged attempt per call. Evidence-first allows 18,000 completion tokens for extraction and 14,000 for planning. CASEPATH allows 10,000 per guard batch; Direct and Graph as context allow 14,000 per call. The charges exclude source preparation, and computation differs across arms.

Product parity. The verifier loads the installed, hash-bound method pack and the product service, then replays all 72 units with the recorded guard decisions. It compares verdicts, document sets and next actions, checks all 36 pair deltas through the product’s diff surface, and verifies that every rendered quotation occurs exactly in its source passage. These comparisons establish implementation identity; an HTTP success code or a plausible demonstration alone would not do so.

Error-origin partition. The retrospective partition forms the union of reference documents from unit annotations and all acceptable signed changes. Each arm retains its original scorer-selected realisation, so its signed-change counts stay unchanged. The bicycle-voucher amendment updated acceptable deltas but left two auxiliary vocabulary fields unchanged. Including both definitions gives the same error counts. Changing one sentence can affect several inferred guards, so the observed absence of withdrawals does not establish monotonicity. The release contains the partition script, all counts and the field-consistency note.

B CORRESPONDENCE BETWEEN THE TWO STUDIES

Both implementations take a source specification, assess case state, plan evidence, replace the state when observations change and derive a justified next action. They differ in knowledge, algorithms, estimands and validation. Study A uses `casepath.theft-method-freeze/5.0.0`, interpreter V4 and evidence overlay V3. Study B uses the compact native producer with `obligation_control_v1` and `evidence_demand_v1`. Its public template union does not transfer the theft source pack, and Study A’s theft pairs are not transformations of the native claim roster.

V5 evaluates local guards, projects all mapped routes for active rules, adds mapped confirmation evidence for true but unconfirmed guards and subtracts held document types. It has no capability-specific or joint adequacy assessment. The native controller separates applicability from acquisition permission and unknown evidence from missing evidence, and selects a route for each capability. The studies make no performance comparison between these runtimes.

Study A’s preparation used model calls to extract propositions, synthesise and refine rules, and map capabilities. The source-only corrections remain in the record, and the held-out cost excludes this preparation. Study B’s compilation is deterministic and shared by every arm.

Binding	Study A: theft-method-freeze/5.0.0	Study B: native-150 compact controller
Knowledge	Source-extracted propositions, macro graph, refined local evidence rules and mapped routes	Common deterministic compilation of all three public tenancy templates
Online gate	Rule-local true/false/unresolved guard; macro edges do not gate requests	Inherited applicability scopes, obligation condition and separate acquisition permission
Evidence state	Held document types; true-guard confirmation; union of mapped routes	Presence, capability/route adequacy, multi-document joint adequacy; per-capability route selection
State update	Recompute requests/next action from a changed packet; paired edit, not acquired evidence outcome	Recompute from a new assessment; benchmark measures one planning realization
Population	9 development + 27 held-out pairs; nine shared concepts, four contexts; 72 units total	60 development/11 families + 90 protected/17 families; 150 claims, three tenancy domains
Learned inference	Guard-major batched V5; 1-call Direct, 1-call representation baseline, 2-call Evidence-first	Five 2-call singleton pipelines; two additional controls reuse CasePath state
Endpoint	Route-aware signed additions/withdrawals, 27-pair micro scores	Complete native artifact and separate actual immediate-checklist metrics
Analysis	Historical family bootstrap, exact family swaps and wrong-family diagnostic; broad gate failed	Equal domain/family/case finite contrasts; family-deletion sensitivity, no significance claim
Parity	Recorded-answer replay over 72 units and 36 deltas	Separate implementation; no inherited V5 or six-role/live-app parity claim

Table 6: A common source-to-state-to-evidence interface does not make the two implementations, populations or measurements interchangeable. Study B extends the evidence question rather than retrospectively re-evaluating V5.

C STUDY B: NATIVE METRICS AND ANALYSIS CONTRACT

Endpoints. Native critical-evidence recall measures criticality-weighted coverage of active evidence obligations through a complete permitted document alternative. Evidence-obligation completeness also requires valid relations from the decision through the fact and capability to the document. Valid-chain precision is the fraction of requested items with such a chain. It is one for an empty request set, so it must be read alongside coverage.

The native contracts define wrong-branch and premature requests, document states, node and edge scores, branch and path probes, and source locators. Locator agreement does not establish semantic entailment; probe accuracy does not directly measure the accuracy of inferred case predicates. The emitted endpoint counts unique immediate request identities without using the reference scenario to activate candidate conditions. It is available only when the reference supplies a unique allowed document set for each active obligation, complete and consistent document states, and the public label mapping.

For predicted and reference sets P, G , checklist $F_1 = 2|P \cap G|/(|P| + |G|)$ and unnecessary fraction $U = |P \setminus G|/|P|$. When both sets are empty, F_1 is one. Empty predictions receive precision one and burden zero; an empty reference receives recall one. Failed cells instead receive the frozen adverse unit penalties: zero for higher-is-better rates and one for lower-is-better rates. Unavailable reference labels cannot be substituted with empty sets or used to reduce a denominator.

Weights and contrasts. For either a split or the complete corpus, scheduled case i receives weight $\omega_i = 1/(3 m_{d(i)} n_{f(i)})$, where m_d is the number of families in its domain and n_f is the number of scheduled cases in its family. Conventional differences use unit scores with adverse imputations. Conservative benefit-oriented differences contribute -1 whenever either producer endpoint is unobserved. The report also deletes each family in turn and renormalises the remaining weights within the

affected domain. The resulting ranges measure sensitivity to corpus composition, without providing confidence intervals.

A comparator counts as beaten only if all three conservative contrasts strictly exceed their boundaries over the complete protected population, using the original references and integrity checks. Otherwise the report records the outcome as not met, a partial practical signal or not evaluable.

Recorded outcomes and failures. Each cell records the arm, case and state, followed by either a validated artefact or the provider output and validation failure. An unknown relation endpoint references a node absent from the candidate artefact. A truncated or non-terminal completion reaches the 4,096-token limit or ends without a terminal outcome. An invalid codebook index fails to resolve in the shared public codebook. A contradictory presence state marks a document as both present and absent, or as present with an inconsistent state. A blocked dependency occurs when a control’s CASEPATH source cell fails. Charges remain attached to every failure, and a repaired version cannot replace a failed cell in the frozen comparison.

D RETROSPECTIVE REQUEST-ONLY DIAGNOSTIC

The registered wrapper scores emitted requests only after a native receipt passes evaluation. The shared expression-namespace mismatch therefore prevents it from scoring any generated request list. The retrospective diagnostic changes this availability rule alone. It reads the unchanged `request_mode=now` items, maps labels through the original public mapping and applies the original emitted-set scorer. It does not evaluate candidate conditions against reference states. The original artefacts, producer and native failures, primary penalties and costs remain available separately.

All 150 references provide an eligible unique document set. The 558 generated lists receive observed request-only scores. The 492 failed or blocked outputs retain the original adverse checklist penalties; no requests are imputed. The original weights for domains, families and cases remain in force. For benefit direction $b_m \in \{-1, 1\}$, the conventional paired contribution is $b_m(s_{i,CP,m} - s_{i,a,m})$. The conservative contribution uses this value when both request-only endpoints are observed and -1 otherwise. A native graph failure does not make a separately scoreable request list unobserved. This diagnostic applies no practical threshold, significance test or confidence interval.

Condition names and their meanings. The public compiler introduces 50 condition symbols used by all arms; every emitted symbol belongs to that namespace. The native expression language evaluates them against the contract’s scenario, which does not supply the required assignment. A target-blind authored counterexample shows why filling this gap requires a semantic choice. For an absent condition, the product controller returns unresolved while the native parser raises a missing-variable error. Substituting null makes the native Boolean evaluation false, whereas the controller still returns unresolved. Renaming a symbol, supplying a default or injecting a predicted state would change its meaning without an explicit state correspondence. The request-only diagnostic makes none of these changes.

Custody and release. The primary scoring task authenticates both complete prediction inventories before it reads references. The original producer ended without its complete I/O audit, leaving historical non-access to protected targets unestablished. Scoring, the compatibility diagnosis and the request-only analysis have separate records that preserve the original failures. All original target containers were then released as exact bytes under the recorded data license. The released version is a public benchmark and cannot provide a new hidden confirmation.

Offline reproduction. The anonymous artifact contains all 1,050 prediction and failure cells, the exact 150 reference containers, original native evaluator, reporting code, source commitments and request-only wrapper. Offline replay reproduces both primary phase reports, the full primary report, every request-only row and the descriptive report with exact canonical JSON equality. It rescores saved outputs without numerical tolerance; it does not regenerate them or include the remote model checkpoint.

Development: 60 claims						
Pipeline	Lists	Correct/req.	Missed	Cond.	Prec.	Recall
CASEPATH	44	259/275	41	811	0.704	0.635
Direct	19	114/116	16	23	0.313	0.281
Document-first	17	95/103	22	84	0.268	0.233
Process-context	27	163/169	23	15	0.430	0.394
Rule-first	28	171/174	21	11	0.455	0.414
Compiled equivalent	44	259/275	41	811	0.704	0.635
Local-scope ablation	44	259/592	41	560	0.331	0.635

Protected families: 90 claims						
Pipeline	Lists	Correct/req.	Missed	Cond.	Prec.	Recall
CASEPATH	66	384/385	54	1195	0.732	0.645
Direct	30	187/200	14	42	0.312	0.310
Document-first	24	142/144	17	55	0.263	0.238
Process-context	43	270/280	13	16	0.462	0.456
Rule-first	42	261/265	15	25	0.460	0.443
Compiled equivalent	66	384/385	54	1195	0.732	0.645
Local-scope ablation	64	376/853	52	827	0.318	0.627

Combined: 150 claims						
Pipeline	Lists	Correct/req.	Missed	Cond.	Prec.	Recall
CASEPATH	110	643/660	95	2006	0.721	0.641
Direct	49	301/316	30	65	0.312	0.298
Document-first	41	237/247	39	139	0.265	0.236
Process-context	70	433/449	36	31	0.451	0.434
Rule-first	70	432/439	36	36	0.458	0.432
Compiled equivalent	110	643/660	95	2006	0.721	0.641
Local-scope ablation	108	635/1445	93	1387	0.323	0.630

Table 7: Request counts and output coverage. Lists counts artefacts that supply raw counts. Correct/req., Missed and Cond. sum over those artefacts only; Cond. counts conditionally deferred documents. Precision and recall use the full population with failure penalties and the specified weights, so they differ from ratios of the raw counts. Dependent controls reuse the same assessments.

Comparator	Endpoint	Conventional	Conservative
Direct	F_1	+0.367	-0.758
Direct	Unnecessary fraction	+0.419	-0.744
Document-first	F_1	+0.429	-0.788
Document-first	Unnecessary fraction	+0.469	-0.785
Process-context	F_1	+0.219	-0.673
Process-context	Unnecessary fraction	+0.270	-0.657
Rule-first	F_1	+0.227	-0.671
Rule-first	Unnecessary fraction	+0.272	-0.650

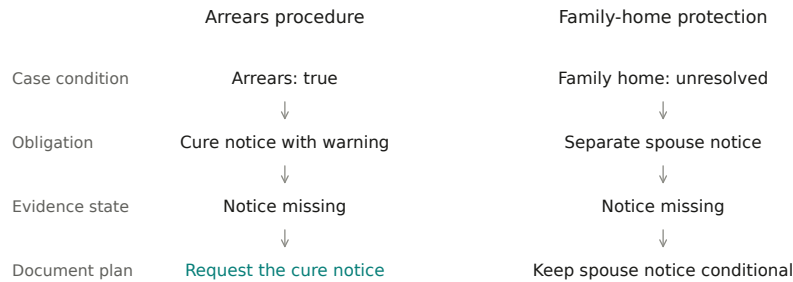
Table 8: Paired request-only contrasts on protected families. Positive values favour CASEPATH. The unnecessary-fraction contrast reverses sign so that a reduction in burden is positive. The conservative contrast assigns -1 when either endpoint is unobserved. The machine-readable report includes every endpoint, family and family-deletion result.

E RETROSPECTIVE EVALUATION AT THE ASSESSED STATE

The evaluated object. The assessed-state interface evaluates each original condition against the complete three-valued assessment saved with its prediction. Known-active entities enter a current-case snapshot; unresolved entities remain recorded but inactive. Counterfactual branch predicates are omitted, excluding branch-probe accuracy and full-program acceptance. The envelope preserves the original artefact, assessment and every activation decision. Predicted activation uses no reference state; native reference graphs, scoring functions and entity matchers are unchanged.

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A missing document becomes a request only when its obligation is active



The recorded next action resolves applicability before acquisition.

Figure 4: A recorded controller decision. The first completed CASEPATH development case in sorted cohort order has true arrears and unresolved family-home status. The controller requests the cure notice immediately and keeps the spouse notice conditional. These are two routes in the saved plan. The exported graph has the same predicate-interface failure as the other generated artefacts. The release provides the full trace and case selection rule.

Negation swaps true and false. Conjunction is true when all children are true and false when any is false; disjunction uses the dual rules. Each control keeps its own graph and requests, bound to its saved CASEPATH assessment. All 110 compiled-equivalent snapshots and scores match CASEPATH, and no immediate request is filtered out. The release includes the derivatives, identity checks and execution receipts.

Metrics and interpretation. Native valid-chain precision counts active requested documents with a reference-aligned decision, fact, capability and document chain as a fraction of all active requests. Critical-evidence recall weights covered reference facts by criticality; obligation completeness counts obligations with valid matching chains. Unnecessary fraction counts requests absent from the reference requirements. Here, valid-chain and necessary documents coincide, making precision and unnecessary fraction complementary. Critical-evidence recall also equals obligation completeness. These are not independent confirmations. Exact-source entailment remains unestablished: provenance-chain inheritance is zero, and omitting predicates makes all-object source support unavailable.

Using per-case benefit when both endpoints exist and -1 otherwise gives chain-precision benefits of $+0.112$ in development, $+0.103$ on protected families and $+0.107$ overall. Critical-recall benefits are -0.261 , -0.289 and -0.278 . These finite-corpus results describe selectivity at comparable coverage. The report gives every family and paired contrast, without post-hoc tests or confidence intervals.

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Endpoint	Development		Protected		Combined	
	Full	Local	Full	Local	Full	Local
Node precision	0.523	0.263	0.536	0.256	0.531	0.258
Node recall	0.542	0.542	0.544	0.526	0.543	0.532
Node F_1	0.530	0.353	0.539	0.344	0.536	0.347
Edge precision	0.676	0.378	0.679	0.368	0.678	0.372
Edge recall	0.374	0.374	0.381	0.370	0.379	0.372
Edge F_1	0.480	0.376	0.487	0.369	0.485	0.371
Critical-evidence recall	0.635	0.635	0.645	0.627	0.641	0.630
Obligation completeness	0.635	0.635	0.645	0.627	0.641	0.630
Valid-chain precision	0.704	0.331	0.732	0.318	0.721	0.323
Unnecessary fraction	0.296	0.669	0.268	0.682	0.279	0.677
Source-chain inheritance	0.000	0.000	0.000	0.000	0.000	0.000
Scored cells	44	44	66	64	110	108
Valid / all requests	259/275	259/592	384/385	376/853	643/660	635/1445

Table 9: Current-case structure and evidence at fixed assessment. Full is CASEPATH with inherited domain scope; Local uses each obligation’s local scope only. Every scheduled case contributes under the original failure penalties and weights for domains, families and cases. Raw counts describe available outputs; coverage differs between controllers on the protected and combined populations. Compiled-equivalent matches Full exactly.